

Colour evidence for a same-redshift quasar companion: the unified 41-band model (October 2026)

Internal method note, `qso_pcolor`

2 October 2026

Abstract

A quasar has a spectroscopic redshift z_0 , and a point source lies a few arcseconds away. This note describes how `qso_pcolor` measures, from broad-band photometry alone, how strongly the companion’s colours support the hypothesis that it is itself a quasar at z_0 . The method compares four populations: quasars at z_0 , quasars at other redshifts, everything else in the imaging catalogue, and a broad catch-all. It accepts any non-empty subset of 41 bands from seven surveys. One latent model, shared by the northern and southern Legacy Surveys, describes all bands. Photometry is corrected for Galactic extinction. Two quasar models are provided: in one, quasar colours do not depend on apparent magnitude (the XDQSO design; the default); in the other, they may. Both are trained on 1.05 million spectroscopic quasars and 1.98 million point sources. They pass all eight quasar-versus-background ranking tests against the previous model. They rank equally well, to ≈ 0.002 in AUC. The magnitude-dependent model describes held-out quasars better by 0.67 nat per object. This note replaces the earlier method note, `docs/method/method.tex`, which describes models retired on 2 October 2026.

Contents

1	Summary of what is delivered	3
2	Notation and glossary	3
2.1	Symbols	3
2.2	Terms	3
3	The question and the ranking statistic	4
4	Data and roles	5
5	Photometry and the observation model	5
5.1	Native luminosities and measurement errors	5
5.2	Galactic extinction	6
5.3	One latent model for Legacy North and South	6
6	The quasar model	7
6.1	Redshift slices and extreme deconvolution	7
6.2	Two variants: with and without magnitude dependence	8
7	The background model	9
8	Fitting	9
8.1	The covariance update	9
8.2	Predictive early stopping	9
9	Priors, catch-all and support	10

10 Validation	10
10.1 Colour–colour distributions	10
10.2 Held-out density	10
10.3 Ranking against the background	11
10.4 Performance across sky, magnitude, redshift and band coverage	11
10.5 Artificial colour grids	12
10.6 Worked examples	12
11 Limitations and open items	12
A The window factorisation	12
B The constrained M step	13

1 Summary of what is delivered

- **Two promoted models.** `models/multisurvey_psf/current` (default): quasar colours independent of apparent magnitude. `models/multisurvey_psf/current_magdep`: quasar colours free to depend on it. `previous` is the 28 September model, kept for rollback. Load with `UnifiedPSFModel.load(...)`.
- **Inputs.** Any non-empty subset of 41 bands (SDSS *ugriz*; Legacy Surveys DR9 South and North *g, r, z, W1, W2*; AllWISE *W1–W4*; Pan-STARRS1 *grizy*; NSC *ugrizY+VR*; SkyMapper *uvgriz*; VHS *YJHK_s*), as catalogued fluxes or magnitudes with errors; candidate RA and Dec; the primary’s redshift z_0 ; morphology and blend information.
- **Output.** A ranking statistic $\log \mathcal{R}$ (Section 3), evidence and posterior quantities, a quasar-support percentile, and explicit eligibility and quality flags.
- **Main results.** Section 10. Both models beat or match the previous model in all eight ranking panels. On artificial colour grids, the failures of earlier candidates (up to 79 spurious high-quasar points) drop to 0 (magnitude-dependent) and 1 (magnitude-independent). Probability calibration is not yet established.

2 Notation and glossary

2.1 Symbols

Symbol	Meaning (units)
z_0	spectroscopic redshift of the primary quasar
z	redshift of a hypothetical companion quasar
$\mathbf{y}$	observed native luptitudes of a source in the bands it has (mag; Section 5)
$\mathbf{C}$	measurement covariance of $\mathbf{y}$ (mag^2 ; diagonal in practice)
$\mathbf{x}$	latent, noise-free luptitudes in the 36 shared coordinates (mag)
$H, \mathbf{b}, T$	observation operator: $\mathbf{y} = H\mathbf{x} + \mathbf{b} + \boldsymbol{\epsilon}$, with extra instrumental scatter $\boldsymbol{\epsilon} \sim N(0, T)$
m_{ref}	luptitude of the reference band, the band on which colours are conditioned (mag)
a	index of the reference band
E	SFD98 reddening $E(B - V)$ at the source position (mag)
R_j	extinction coefficient of band j , A_j/E
ℓ, b	Galactic longitude and latitude (deg)
$\Sigma_Q(z, m)$	quasar surface density per unit redshift and magnitude ($\text{deg}^{-2} \text{mag}^{-1}$)
$\Sigma_B(m)$	background surface density per unit magnitude ($\text{deg}^{-2} \text{mag}^{-1}$)
η	catch-all share of the background intensity (dimensionless)
λ_X	intensity of hypothesis X : surface density times colour likelihood
$\mathcal{R}$	ranking statistic, Eq. (3) (per unit redshift)
Δz_{eff}	effective width of the declared redshift window, <code>dz_match_eff</code>
K	number of Gaussian components in a mixture
w, ν	covariance-prior scale (10^{-3}mag^2) and strength (1)

2.2 Terms

Luptitude. The inverse hyperbolic sine magnitude of Lupton et al. (1999),

$$m = 22.5 - \frac{2.5}{\ln 10} \left[\sinh^{-1} \left(\frac{f}{2s} \right) + \ln s \right],$$

for flux f in the band’s native unit and softening s . It equals the ordinary magnitude for $f \gg s$ and stays finite for zero or negative flux.

Native system. A band’s own catalogue calibration. No band is converted into another survey’s system.

Slice. The quasar model for redshifts within ± 0.1 of a centre z_k , for $z_k = 0.15, 0.25, \dots, 4.35$ (43 slices; neighbouring slices overlap).

Reference band. The first band, in a fixed survey-priority order, that the source has observed at signal-to-noise ≥ 5 . Colour likelihoods are conditioned on it.

Role. Every training object belongs to one sky-based role: *fit* and *select* (training), *calibration* (catch-all shares and support threshold), or *test* (validation). Roles are fixed in advance and never reassigned.

Background. Every point source in the imaging catalogue that is not a known quasar. It is dominated by stars, and it is fitted empirically rather than as a pure stellar model.

Catch-all. A broad Student- t density with a small share of the background intensity. It explains objects that neither the quasar nor the background mixture fits, so that Gaussian tails do not decide their classification.

Support percentile. For a candidate and z_0 , the fraction of simulated quasars at z_0 , with the candidate's own errors, whose conditional density is lower than the candidate's. A small value means the colours are unusual for a quasar at z_0 .

Held-out density. The mean, over test objects, of the conditional log density $\ln p(\mathbf{y}_{\text{other}} \mid m_{\text{ref}})$ (nat per object, density per mag^{N-1}). Higher is better.

AUC. The probability that a random test quasar ranks above a random test background object on $\log \mathcal{R}$. Objects the scorer declines to rank are placed at the bottom.

3 The question and the ranking statistic

Four hypotheses compete for a companion with photometry $\mathbf{y}$ near a primary at z_0 (Fig. 1):

1. $Q(z_0)$, a quasar at the primary's redshift, within a declared window;
2. $Q(z \neq z_0)$, a quasar at any other redshift;
3. B , a member of the background population;
4. U , the catch-all.

Each hypothesis has an intensity, a surface density times a likelihood. With reference magnitude m and conditional colour likelihoods,

$$\lambda_Q(z) = \Sigma_Q(z, m) p(\mathbf{y} \mid Q, z, m), \quad (1)$$

$$\lambda_B = (1 - \eta) \Sigma_B(m) p(\mathbf{y} \mid B, m, \ell, b), \quad \lambda_U = \eta \Sigma_B(m) p(\mathbf{y} \mid U, m). \quad (2)$$

We rank candidates on

$$\mathcal{R} = \frac{\lambda_Q(z_0)}{\int \lambda_Q(z) dz + \lambda_B + \lambda_U} \quad [\text{per unit redshift}]. \quad (3)$$

The denominator does not depend on the redshift window, because same-redshift and other-redshift quasars together form the whole quasar intensity. A posterior for a window of effective width Δz_{eff} follows by one multiplication, $p_{\text{same}} = \mathcal{R} \Delta z_{\text{eff}}$ (exact for the window-averaged $\mathcal{R}$ that the scorer stores). Photometric redshifts have widths of about $\sigma_z \approx 0.6$, while a $\pm 2000 \text{ km s}^{-1}$ window at $z = 1.4$ is $\Delta z = 0.016$ wide, so p_{same} is small even for an ideal candidate. $\log \mathcal{R}$ is the quantity to compare between candidates. Appendix A gives the derivation.

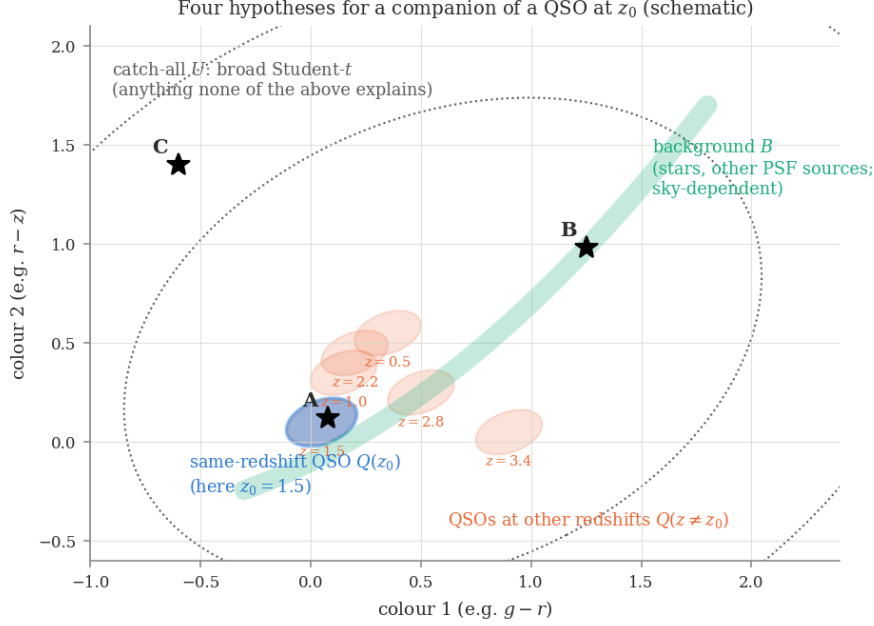


Figure 1: Schematic (synthetic, not fitted) of the four hypotheses in one colour plane. Candidate A lies on the same-redshift quasar locus, B on the background locus, and C far from both; the catch-all explains C, so the Gaussian tail of whichever population is wider does not decide it.

4 Data and roles

Quasars. The training quasars are the union of DESI DR1 quasars (`spectype=QSO`, `zwarn=0`, primary) and SDSS DR16Q (Lyke et al., 2020), merged by sky position. Of 1,657,444 eligible quasars, 1,049,260 are fit or select rows. Of these, 610,503 are DESI-only, 241,808 SDSS-only, and 196,949 in both (Fig. 3). The remaining quasars are 239,371 calibration and 368,813 test rows. Their photometry was acquired from all seven surveys and stored as native fluxes and variances.

Background. The background sample is 2,978,857 cleaned point sources in 274 sky regions (Legacy `type=PSF`, mask-clean, known quasars removed), with seven-survey photometry. Of these, 1,975,894 are fit or select rows and 502,251 calibration rows.

Selection. All samples satisfy $|b| \geq 25^\circ$. A band is used only when its survey marks it usable (finite flux, positive variance, a positive exposure count where reported, and the survey’s cleanliness flag); weak and negative fluxes are kept. No signal-to-noise cut is applied to training. Validation statistics use objects with at least one band at signal-to-noise ≥ 5 .

Roles. Roles are assigned by sky region before any fitting (Fig. 2). Rows used during development to choose stopping points or thresholds (34,259 quasars and 2,045 background objects) are recorded and excluded from every test reported in Section 10.

5 Photometry and the observation model

5.1 Native luptitudes and measurement errors

Every band keeps its native calibration and zero point (AB for SDSS, Legacy, PS1, NSC and SkyMapper; Vega for AllWISE and VHS). Fluxes become luptitudes with a per-band softening s , and their variances propagate through the exact derivative of the luptitude function. A band that is not observed is marginalised exactly; it is never imputed.

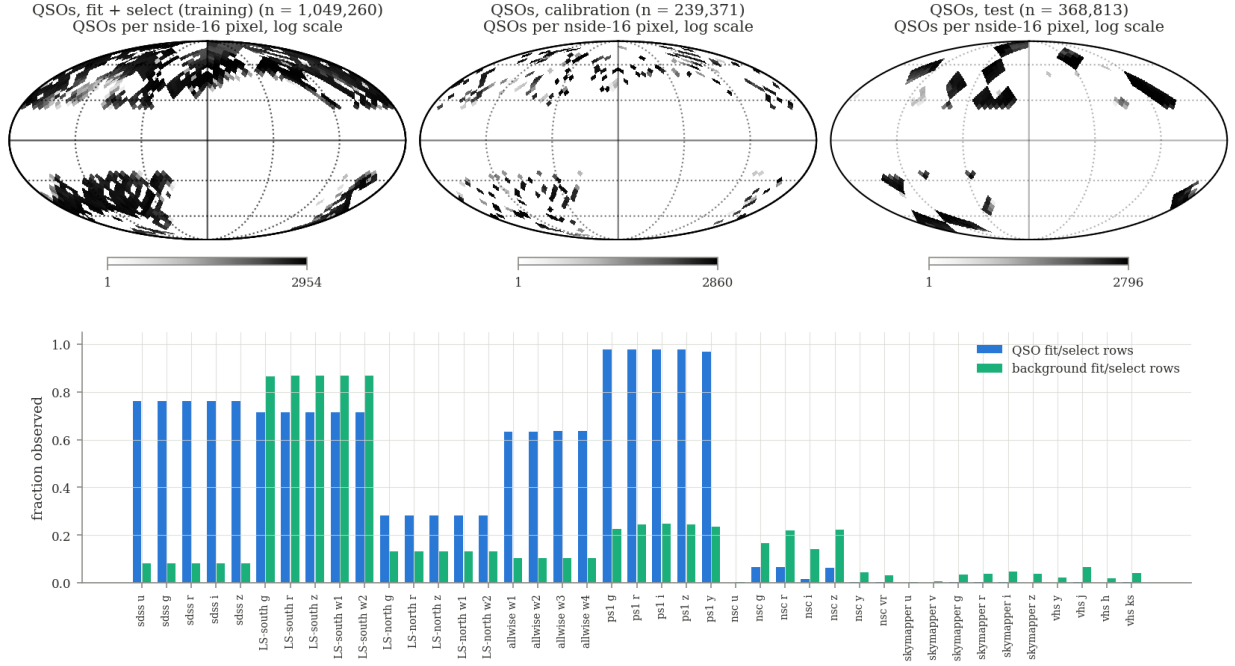


Figure 2: Top: sky distribution of quasars by role, Galactic Mollweide projection, counts per HEALPix nside-16 pixel (log scale). Bottom: fraction of fit/select rows with each band observed, for quasars (blue) and the background (aqua).

5.2 Galactic extinction

Fluxes and variances are corrected before any other step:

$$f \rightarrow f 10^{0.4R_j E}, \quad \sigma_f^2 \rightarrow \sigma_f^2 10^{0.8R_j E}, \quad (4)$$

where E is the SFD98 reddening (Schlegel et al., 1998) at the source position. This preserves signal-to-noise and keeps negative fluxes negative. Table 1 lists the coefficients. Legacy DR9 values come from the survey documentation (Legacy Surveys, 2020), SDSS and PS1 values from Schlafly & Finkbeiner (2011), Table 6, and SkyMapper values from Wolf et al. (2018). NSC VR and VHS Y have no published values in these sources. We computed them with the Schlafly & Finkbeiner (2011) method: an F99 law with $R_V = 3.1$ and a 7000 K source, with one scale factor fitted to 16 published rows (residual rms 0.73%). Top-hat bandpasses were used, so these two values are approximate; we estimate $\pm 5\%$ for VR . The scorer computes E from the candidate’s position and applies the same correction, so users supply catalogue fluxes as observed. Fig. 4 shows that our correction reproduces the Legacy catalogue correction (`mw_transmission`) to 0.002 mmag.

5.3 One latent model for Legacy North and South

Legacy DR9 South (DECam) and North (BASS/MzLS) g, r, z use different filters. Earlier models fitted them as separate bands, and the northern fits then borrowed little from the much larger southern sample. We now describe every source by latent, noise-free luminosities $\mathbf{x}$ in 36 coordinates: five shared Legacy coordinates (`legacy:g,r,z,W1,W2`) and 31 coordinates for the other surveys. The 41 native bands follow from a linear operator,

$$\mathbf{y} = H\mathbf{x} + \mathbf{b} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim N(0, T). \quad (5)$$

Rows of H for the southern Legacy and all non-Legacy bands are unit vectors. Rows for the northern g, r, z carry the DESI North→South colour relation. For example, $y_{g,N} = 1.0627 x_g - 0.0643 x_r + 0.0016 x_z - 0.0060$. T holds the instrumental scatter measured on overlap pairs. The

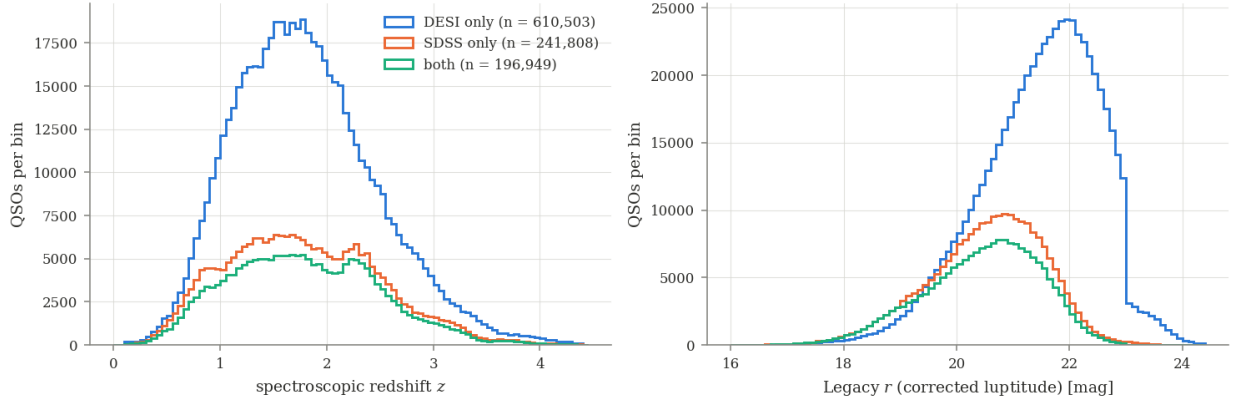


Figure 3: Fit/select quasars by parent survey. Left: spectroscopic redshift. Right: Legacy r (extinction-corrected luptitude, southern band where observed, otherwise northern). The DESI-only sample ends near DESI’s quasar target limit, $r \approx 23$.

Table 1: Extinction coefficients $R_j = A_j/E(B - V)_{\text{SFD}}$.

Bands	R_j	Source
Legacy DECam g, r, z	3.214, 2.165, 1.211	Legacy DR9 documentation
Legacy North g, r, z	3.278, 2.189, 1.196	DECam values through the DESI relation, Eq. (6)
$W1, W2$ (Legacy, AllWISE); $W3, W4$	0.184, 0.113; 0.0241, 0.00910	Legacy DR9 documentation
SDSS $ugriz$	4.239, 3.303, 2.285, 1.698, 1.263	Schlafly & Finkbeiner (2011), $R_V = 3.1$
PS1 $grizy$	3.172, 2.271, 1.682, 1.322, 1.087	Schlafly & Finkbeiner (2011)
NSC $ugrizY$	3.995, 3.214, 2.165, 1.592, 1.211, 1.064	DECam (Legacy DR9)
NSC VR	2.28	computed, approximate
SkyMapper $uvgriz$	4.294, 4.026, 2.986, 2.288, 1.588, 1.206	Wolf et al. (2018)
VHS $J, H, K_s; Y$	0.709, 0.449, 0.302; 1.00	UKIRT rows of Schlafly & Finkbeiner (2011); Y computed

relation acts on uncorrected magnitudes, so extinction is removed in the DECam system:

$$R_{\text{native}} = H R_{\text{latent}}, \quad (6)$$

where each latent coordinate takes the coefficient of the system it represents. Fig. 5 shows that the relation removes the colour-dependent North–South offsets on the same objects. The Legacy $W1, W2$ coordinates are shared after both hemispheres use common units and softening; they remain distinct from AllWISE $W1, W2$.

6 The quasar model

6.1 Redshift slices and extreme deconvolution

For each of the 43 slices, the latent quasar luptitudes follow a Gaussian mixture with K components ($K = 20$ for 23 slices, 12 for 6, 8 for 13, and 4 for the sparsest one, 640 objects at $z = 4.35$):

$$p(\mathbf{x} \mid Q, z_k) = \sum_{i=1}^K \alpha_i N(\mathbf{x}; \boldsymbol{\mu}_i, V_i). \quad (7)$$

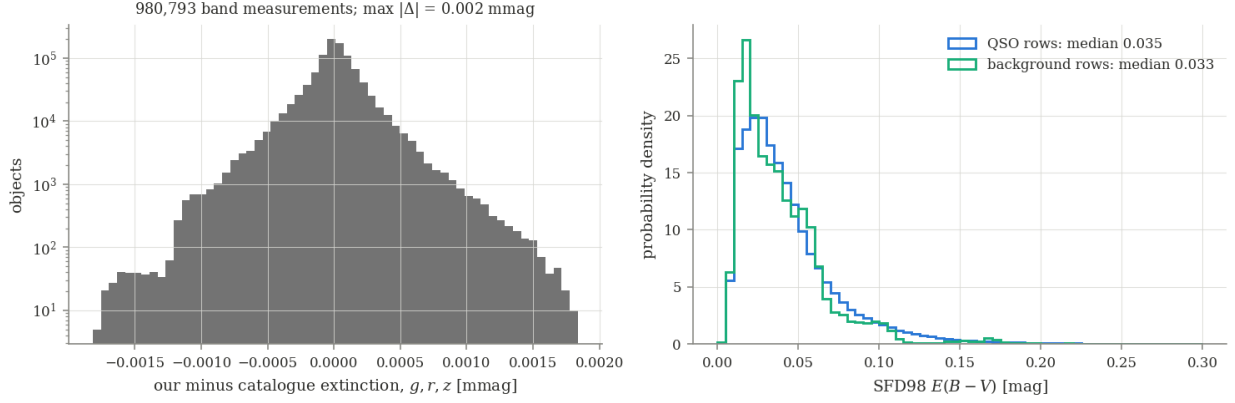


Figure 4: Left: our correction minus the Legacy catalogue correction, in four reserved Legacy cones (980,793 g, r, z measurements). Right: distribution of SFD98 reddening for the quasar and background rows (medians 0.035 and 0.033 mag).

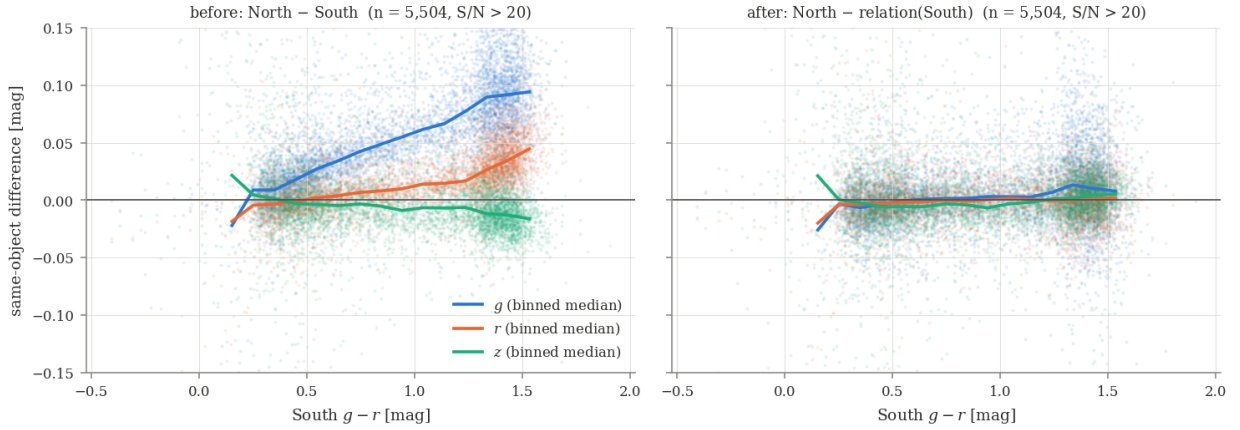


Figure 5: Same-object North minus South luminosities in overlap fields ($S/N > 20$ in all six bands), against South $g - r$. Left: raw difference. Right: after applying the DESI relation to the southern values. Lines: binned medians.

Extreme deconvolution (Bovy et al., 2011a) fits α_i, μ_i, V_i to noisy, incomplete data. Each object contributes $N(\mathbf{y}; H\mu_i + \mathbf{b}, HV_iH^T + T + \mathbf{C})$ in its observed bands only, so measurement errors are removed from the fitted widths rather than absorbed into them. Section 8 describes the fitting. At scoring time the candidate’s own covariance is added back, and the density is conditioned on the reference band: $p(\mathbf{y}_{\text{other}} | m_{\text{ref}}) = p(\mathbf{y})/p(m_{\text{ref}})$. The scorer interpolates linearly between neighbouring slices in redshift.

6.2 Two variants: with and without magnitude dependence

Conditioning a joint mixture on m_{ref} lets a model’s colour distribution depend on apparent magnitude: a component’s colour mean shifts with m_{ref} unless its covariance forbids it. The XDQSO design (Bovy et al., 2011b) forbids it, and leaves all magnitude dependence to the abundance prior $\Sigma_Q(z, m)$. We provide both:

Magnitude-independent (default). We rewrite the latent vector as one magnitude and 35 colours, $u_r = x_r$ and $u_j = x_j - x_r$ for $j \neq r$, with r the shared Legacy r coordinate. In every component the magnitude coordinate has the same fixed distribution, $N(20, 100 \text{ mag}^2)$, and zero covariance with the colours. Conditioning on any reference band then leaves the colour likelihood independent of brightness. The residual dependence through the finite magnitude variance is below

10^{-3} nat for a ± 3 mag shift. With this block structure the likelihood factorises, and the exact constrained M step is the ordinary update of the colour block. A unit test confirms that it equals a fit to the colours alone.

Magnitude-dependent. The same mixture without the constraint.

Why both are kept. DESI and SDSS selected their quasars with colour cuts that change with magnitude. A model allowed to depend on magnitude can learn that selection as if it were quasar physics, and then penalise genuine quasars outside the selection. The test quasars share the training selection, so our tests cannot measure this risk. They can measure the cost of the constraint, which Section 10 reports.

7 The background model

The background is one Gaussian mixture with $K = 20$ components over the same 36 latent coordinates, fitted by extreme deconvolution to all 1,975,894 fit/select background rows. Its colours depend on magnitude through the reference-band conditioning. They depend on sky position through HEALPix-dependent component weights: shared Gaussian shapes, with weights fitted per inside-4 cell and shrunk towards inside-2 parent cells and the all-sky weights. The pooling count is $n_0 = 1000$. The surface density $\Sigma_B(m)$ is measured per reference band from counts in regions of measured usable area, pooled with $n_0 = 100$. A candidate-local refit of weights and density in a declared cone is available.

8 Fitting

8.1 The covariance update

The extreme-deconvolution E step computes each component’s responsibility q_{ni} for object n , and the posterior moments of $\mathbf{x}$. The M step updates the weights, the means, and the covariance from the scatter S_i about the new mean. A previous implementation used $V_i = S_i + wI$. That step maximises no fixed objective, because its penalty is weighted by the changing responsibilities. In 24 of 43 earlier full-data quasar fits the likelihood therefore declined, by up to 0.078 nat per object below its best value, and a stopping rule that required a non-negative change could never stop such fits. We now use the conjugate-prior MAP update of Bovy et al. (2011a) (their eqs. 19–20):

$$V_i = \frac{q_i S_i + \nu w I}{q_i + \nu}, \quad q_i = \sum_n q_{ni}, \quad \nu = 1, \quad w = 10^{-3} \text{ mag}^2, \quad (8)$$

which maximises the log posterior with prior $-\frac{\nu}{2} [\ln |V_i| + w \text{tr}(V_i^{-1})]$ per component. EM therefore cannot decrease the reported objective (Fig. 7a).

8.2 Predictive early stopping

Every 20 updates we compute the held-out density on a fixed stopping panel: up to 512 test-role objects per hemisphere per fit, recorded and excluded from later validation. A fit stops when a block gains less than 0.02 nat per object, when the objective changes by less than 10^{-4} (relative), or after 300 updates. The best checkpoint, including the warm start, is kept. The 0.02 nat threshold is about twice the sky-bootstrap standard error of late block gains. This is predictive early stopping, not convergence. Sparse slices benefit most: where held-out density falls in the first block, as at $z = 4.05$, the fit keeps its warm start. All 44 fits of each production model stopped this way, with no objective decrease (Fig. 7c,d).

9 Priors, catch-all and support

Abundance priors. $\Sigma_Q(z, m)$ and $\Sigma_B(m)$ are stored per reference band, per unit native luptitude. Background densities are recounted from corrected photometry. The quasar abundance is inherited from the previous model and moved onto corrected magnitudes by the mean extinction shift: 0.080 mag in Legacy r , at mean $E = 0.037$. This is an approximation, not a recount.

Catch-all. U is a Student- t with $\nu = 4$ degrees of freedom, centred on the background mean, with the background mixture’s second-moment matrix as its scale. Measurement noise is convolved exactly through the t ’s Gaussian scale-mixture representation. Its share η is a Beta-MAP fit per reference band and three magnitude bins, on the 502,251 calibration background objects.

Support cut. For a candidate, z_0 and the observed bands, the scorer draws 255 simulated quasars from the conditional model at z_0 , adds the candidate’s errors, and ranks the candidate’s conditional density among them. Candidates below a threshold are not ranked; their likelihoods are kept as diagnostics. The threshold is set on 800 calibration quasars (four band subsets, 3,027 evaluations). At the promoted 99.5% target it is $2/256 = 0.0078$, which retains 99.7% of calibration evaluations. The target was chosen on the release test rows (Fig. 8); Section 10.4 confirms it on independent rows.

10 Validation

Tests use test-role objects outside every recorded development exclusion. “Previous” denotes the 28 September model.

10.1 Colour–colour distributions

Figures 9–11 compare model densities with the data in two colour planes: Legacy South ($g-r, r-z$) and ($r-z, z-W1$). Model contours are conditioned on each bin’s median r , weighted by the data’s redshift mix, and convolved with the bin’s median measurement covariance, so they can be compared directly with the data. Each panel title gives the fraction of the data inside the model’s 50% and 90% regions; a well-calibrated model gives 0.50 and 0.90. The magnitude-dependent quasar model gives 0.48–0.51 and 0.89–0.90 in every magnitude bin and both planes. The magnitude-independent model must use one width for all magnitudes. It is too broad for bright quasars (0.62/0.94 at $17 \leq r < 19$) and too narrow for faint ones (0.41/0.86 at $21.5 \leq r < 22.5$). The apparent colour scatter of quasars, after deconvolution, grows towards faint magnitudes in this sample. Underestimated photometric errors, a change in the population, and selection can all produce this. The background model is less well calibrated: its 50% coverage ranges from 0.36 to 0.62. Its broad outer contours come from low-density tails spread over empty colour space, not from missing the stellar locus.

10.2 Held-out density

Per slice, the magnitude-dependent model’s kept checkpoint has a held-out density higher than the magnitude-independent model’s by a median 0.666 nat per object (range 0.381–0.876). Against the uncorrected 30 September model, the corrected magnitude-dependent model differs by a median +0.001 nat (range -0.262 to $+0.057$). The extinction correction therefore costs essentially nothing in density, and the density gap between the two promoted models comes from the magnitude constraint.

Table 2: Quasar-versus-background AUC on release test rows (500 per class per hemisphere), with the support cut at 99.5% retention. “Panel” gives the bands supplied to the scorer and the hemisphere.

Panel	previous	magnitude-independent	magnitude-dependent
all bands, South	0.945	0.968	0.980
all bands, North	0.941	0.982	0.980
Legacy g, r, z , South	0.978	0.987	0.985
Legacy g, r, z , North	0.986	0.987	0.989
SDSS, South	0.976	0.977	0.981
SDSS, North	0.967	0.979	0.978
PS1, South	0.941	0.937	0.938
PS1, North	0.952	0.955	0.955

10.3 Ranking against the background

Table 2 gives the quasar-versus-background AUC in the eight release panels, with the support cut at 99.5%. Both models are within the 0.005 tolerance of the previous model, or better, in every panel.

10.4 Performance across sky, magnitude, redshift and band coverage

These tests use larger panels: 20,000 quasars and 20,000 background objects per hemisphere. The rows are test-role and lie outside every development exclusion and outside the release rows that chose the support threshold (35,259 quasars and 3,045 background objects excluded in total). Background objects are scored at redshifts drawn from the test quasars’ redshifts. The four metrics are those of Section 2: AUC, held-out density, support retention, and high-QSO incidence, the fraction of background objects ranked with $p_{\text{quasar}} > 0.5$.

Overall, and an independent check of the support cut. On these rows, the cut keeps 99.45% (magnitude-independent) and 99.50% (magnitude-dependent) of test quasars. Both values agree with the 99.5% calibration target, so the threshold carries over to rows that did not choose it. The overall AUC is 0.973 and 0.978 (South 0.971 and 0.975; North 0.976 and 0.980), and the high-QSO incidence among background objects is 0.44% and 0.46%.

Sky position and magnitude (Fig. 12). Per-cell AUC is uniform across the footprint to within a few thousandths. Against $|b|$, AUC changes little (0.970–0.978 magnitude-independent, 0.975–0.984 magnitude-dependent), and it is highest at the lowest latitudes, where the background is most stellar. Against reference magnitude, the two models differ most for bright quasars: AUC 0.959 against 0.976 at $r < 18$ (251 quasars), and 0.981 against 0.992 at $18 \leq r < 19$. Here the magnitude-independent model’s single colour width is too broad (Section 10.1). Both models converge at $r > 21$. High-QSO incidence peaks at 1.4–1.9% for $20 \leq r < 22.5$. These are random point sources, so the incidence is not a contamination rate: some of them are unrecognised quasars.

Redshift (Fig. 13). AUC is 0.965–0.981 for $0.5 \leq z_0 \leq 3$, and falls to 0.94 at $z_0 = 3.45$, where quasar colours cross the stellar locus. Held-out density falls with redshift as quasars become fainter, and the magnitude-dependent model stays higher at every redshift. Support retention is above 99% everywhere.

Band coverage (Fig. 14). Grouping objects by the surveys that actually observed them mixes coverage with brightness and redshift; for example, objects with SDSS photometry are brighter. Some groups also contain fewer than 150 background objects, which makes their AUCs noisy. The artificial masks remove both problems, because they score the same objects with fewer bands. With Legacy g, r, z only, AUC is 0.986 and 0.987; with all Legacy bands, 0.994 and 0.995; with SDSS

only, 0.975 and 0.976; with PS1 only, 0.957 and 0.960; with all optical bands, 0.990 for both. The largest difference between the models is for objects with Legacy South photometry alone (AUC 0.955 against 0.977, 328 quasars). These are faint DESI-only quasars.

10.5 Artificial colour grids

A colour grid places synthetic sources on a 15×15 lattice in $(g-r, r-z)$ from -6 to 8 mag at fixed r . These are stress tests of extrapolation into empty colour space, not contamination rates. In the original low-density region, an unpromoted 30 September candidate gave 72 and 79 high-quasar points ($p_{\text{quasar}} > 0.5$) in the North, and the previous model gave 12 at $r = 21$. With the support cut at 99.5%, the magnitude-dependent model gives none and the magnitude-independent model one, at South $r = 18.5$ (Fig. 15). They abstain from most of the grid: points outside both models, or rejected by the support cut, are not ranked. Without the support cut, the magnitude-independent model would give 28 high-quasar grid points and the magnitude-dependent one 6, so the default model relies on the cut more.

10.6 Worked examples

Figure 16 shows six test objects scored by both models. For each band the panels compare the observed corrected magnitude, minus the reference band, with the model’s conditional prediction at the object’s redshift. A bright $z = 1.51$ quasar and a $z = 0.65$ quasar match their predictions and receive $p_{\text{quasar}} = 1.00$. A $z = 3.04$ quasar with noisier photometry receives a lower $\log \mathcal{R} \approx -0.5$, though still $p_{\text{quasar}} > 0.9$. One test quasar has red optical colours, far from the $z = 1.5$ locus, and both models decline to rank it. A background star has $\log \mathcal{R} < -40$. The last panel is a background object that both models call a quasar ($p_{\text{quasar}} = 0.65$ and 0.86). Its colours lie on the quasar locus, and it may be an unrecognised quasar. Legacy and AllWISE $W1, W2$ appear as separate points because Legacy fluxes are AB and AllWISE Vega; the zero points differ by about 2.7 and 3.3 mag.

11 Limitations and open items

- **Probability calibration** is not established. AUC and density measure ranking and fit, not the reliability of p_{same} or p_{quasar} .
- **Selection.** Test quasars share the training selection. Our tests cannot measure the intended benefit of the magnitude-independent model for quasars outside DESI and SDSS selection.
- **Quasar abundance** is inherited and shifted, not recounted.
- **Sparse surveys.** NSC VR and VISTA Y extinction coefficients are approximate, and SkyMapper u, v and VHS have little training support.
- **Background calibration** in colour-colour projections is moderate (Section 10.1).
- **Out of scope.** Blends below the separation floor, and a physical-pair clustering prior, are deferred.

A The window factorisation

Write $\Lambda_Q = \int \lambda_Q(z) dz$ for the total quasar intensity, and let the declared window W around z_0 have effective width $\Delta z_{\text{eff}} = \int_W k(z) dz$ for its kernel k . The posterior that the companion is a quasar in the window is

$$p_{\text{same}} = \frac{\int_W k(z) \lambda_Q(z) dz}{\Lambda_Q + \lambda_B + \lambda_U} = \Delta z_{\text{eff}} \frac{\langle \lambda_Q \rangle_W}{\Lambda_Q + \lambda_B + \lambda_U} \equiv \Delta z_{\text{eff}} \mathcal{R}, \quad (9)$$

with $\langle \lambda_Q \rangle_W$ the kernel-weighted mean over the window. The denominator contains no window, because Λ_Q includes both same-redshift and other-redshift quasars. The scorer integrates the window on dense nodes at the support boundaries rather than on the coarse redshift grid. For a window much narrower than the photometric redshift width, $\langle \lambda_Q \rangle_W = \lambda_Q(z_0)$, which is Eq. (3).

B The constrained M step

For a component with mean $(\mu_m, \boldsymbol{\mu}_c)$ and covariance $\text{diag}(\sigma_m^2, V_c)$, where μ_m and σ_m^2 are fixed, the expected complete-data log likelihood separates into a magnitude term, constant in the free parameters, and a colour term. Maximising the colour term with the MAP prior on V_c gives the usual updates applied to the colour block alone. The prior on the full covariance differs from the prior on V_c by a constant. The constrained update is therefore exact.

References

- Bovy J., Hogg D. W., Roweis S. T., 2011, *Ann. Appl. Stat.*, 5, 1657, “Extreme deconvolution”, [arXiv:0905.2979](#).
- Bovy J. et al., 2011, *ApJ*, 729, 141, “Think outside the color box” (XDQSO), [arXiv:1011.6392](#).
- Lupton R. H., Gunn J. E., Szalay A. S., 1999, *AJ*, 118, 1406, [arXiv:astro-ph/9903081](#).
- Lyke B. W. et al., 2020, *ApJS*, 250, 8, SDSS DR16Q, [arXiv:2007.09001](#).
- Schlegel D. J., Finkbeiner D. P., Davis M., 1998, *ApJ*, 500, 525, [doi:10.1086/305772](#).
- Schlafly E. F., Finkbeiner D. P., 2011, *ApJ*, 737, 103, [doi:10.1088/0004-637X/737/2/103](#).
- Wolf C. et al., 2018, *PASA*, 35, e010, SkyMapper DR1, [doi:10.1017/pasa.2018.5](#).
- Legacy Surveys DR9 catalogue documentation, Galactic extinction coefficients, <https://www.legacysurvey.org/dr9/catalogs/>.

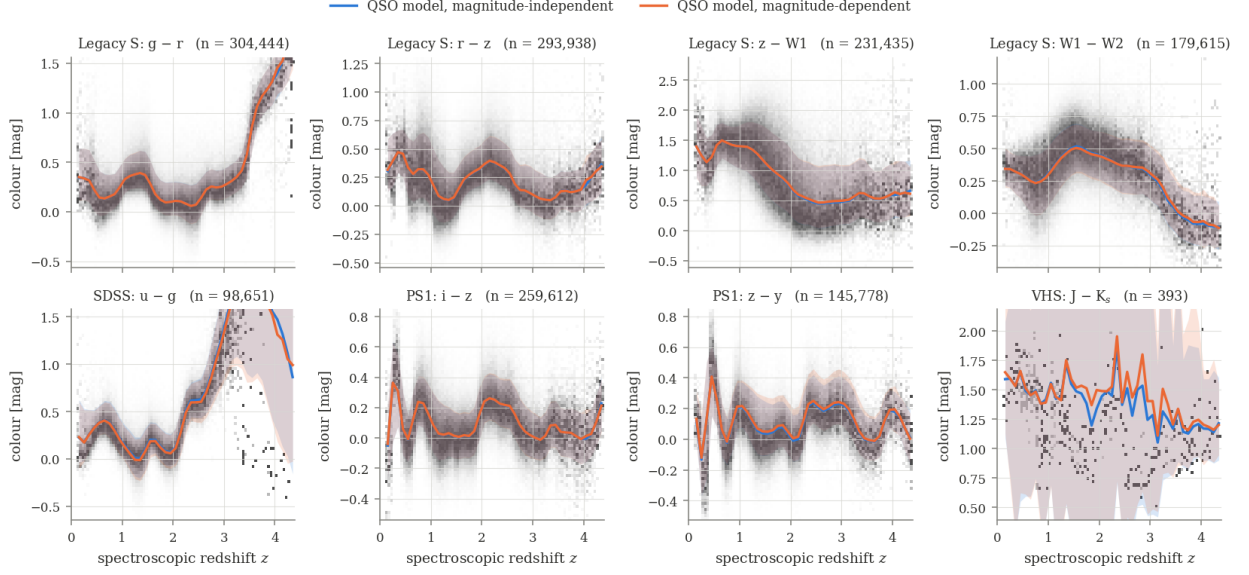


Figure 6: Quasar colour against redshift. Grey: test quasars with both bands at $S/N > 10$ (density normalised in each redshift column of width 0.05). Lines and bands: per-slice median and 16–84% range of the noise-free model colour, magnitude-independent (blue) and magnitude-dependent (orange, marginalised over magnitude). The data include measurement noise and a signal-to-noise selection that the model bands do not.

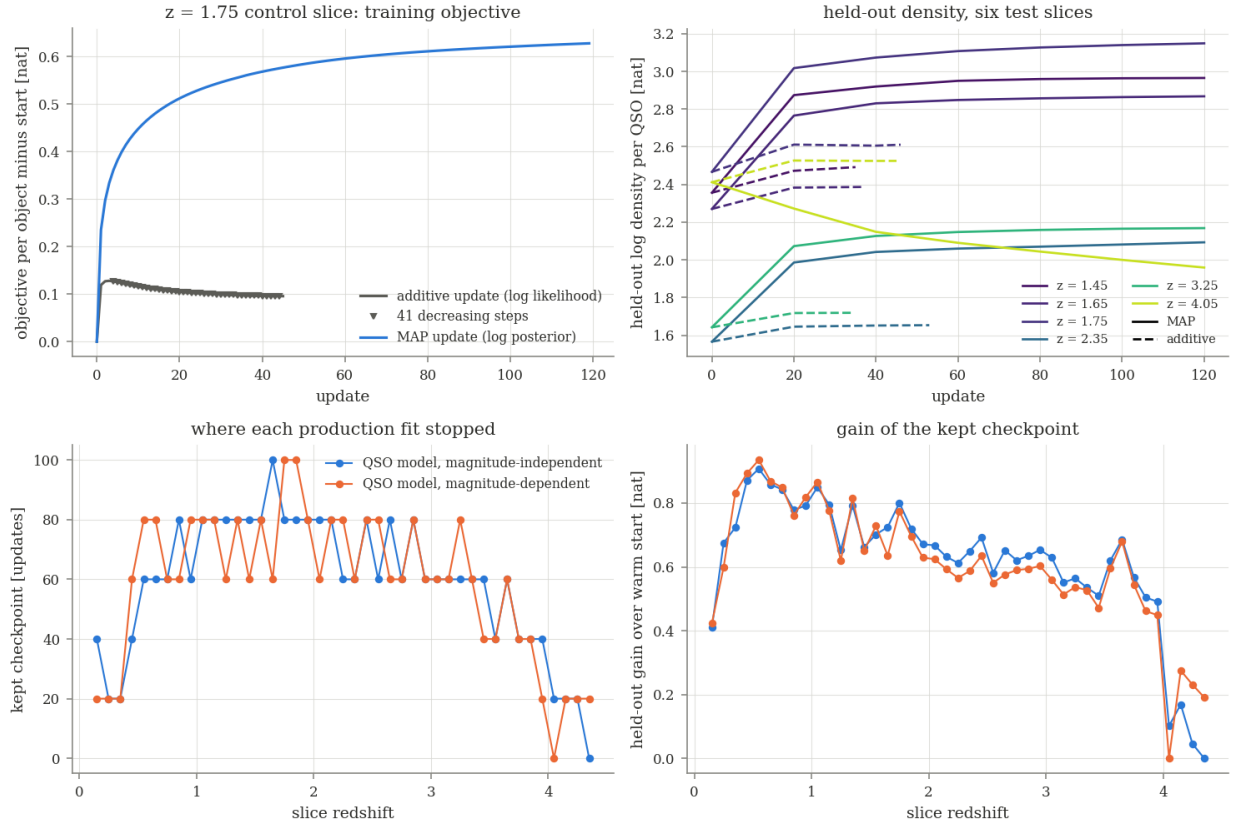


Figure 7: (a) Training objective per object minus its starting value, $z = 1.75$ slice: additive update (log likelihood; 41 decreasing steps) and MAP update (log posterior). (b) Held-out density for six slices under both updates (identical rows and warm starts). (c) Update at which each production fit’s kept checkpoint was taken. (d) Held-out gain of that checkpoint over the model’s own warm start.

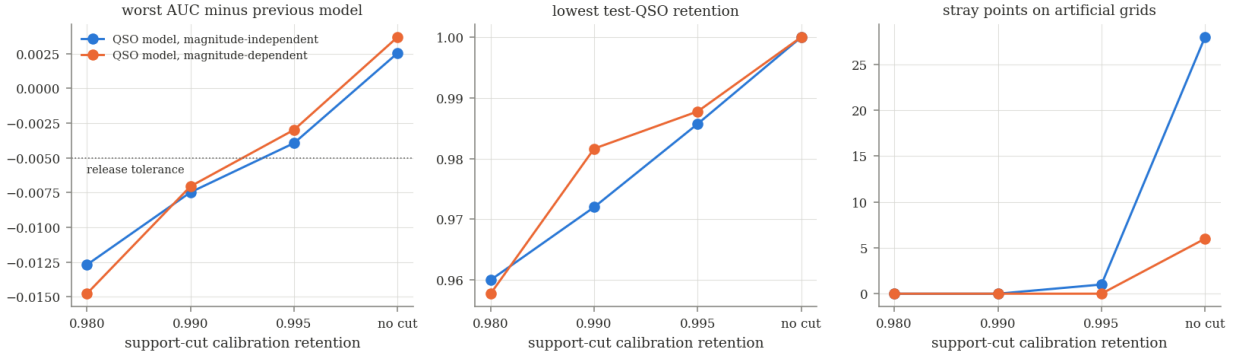


Figure 8: Effect of the support-cut target retention on the release panels: worst AUC difference from the previous model over eight panels, lowest fraction of test quasars kept, and high-quasar points on artificial colour grids (Section 10.5).

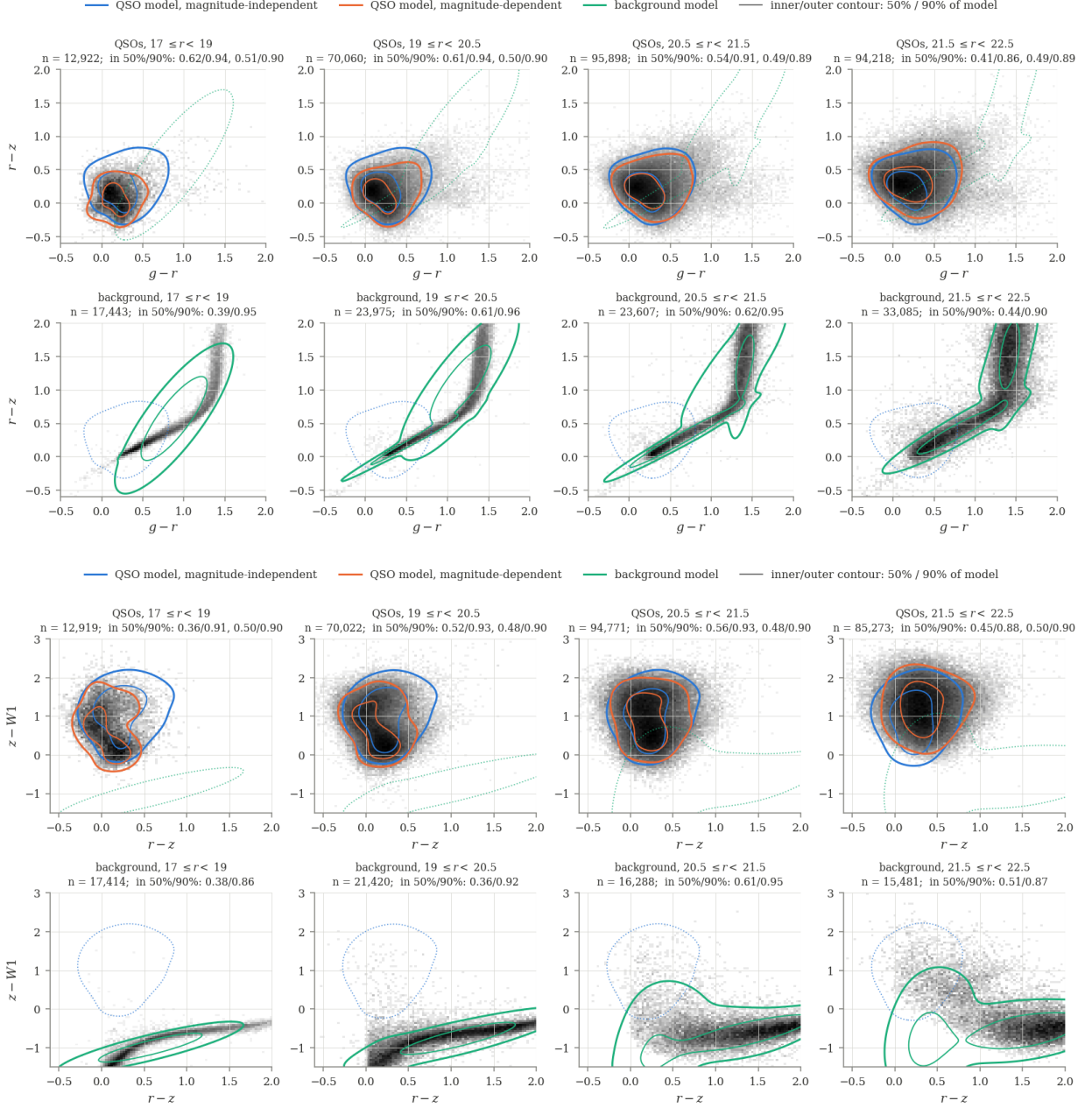


Figure 9: Colour-colour planes in bins of Legacy South r . Top rows of each block: test quasars (grey, log density) with both quasar planes models (blue, orange; thick and thin contours enclose 90% and 50%) and the background model's 90% contour (dotted aqua). Bottom rows: test background objects with the background model and the magnitude-independent quasar model's 90% contour (dotted blue).

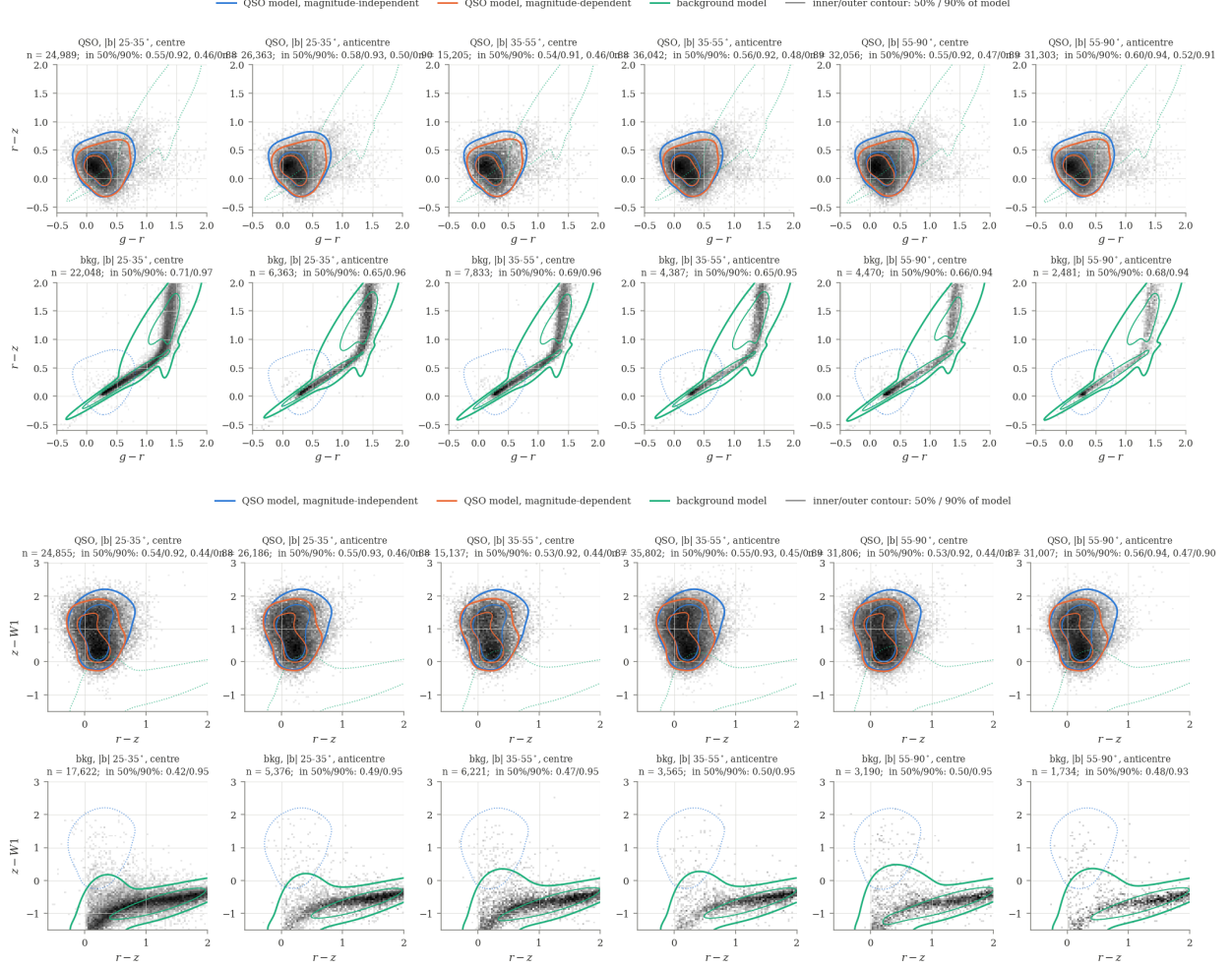


Figure 10: As Fig. 9, in bins of Galactic latitude and of longitude towards ($|b| < 90^\circ$) or away from the Galactic centre, at $19 \leq r < 21.5$. Background contours use the spatial component weights averaged over each bin's objects.

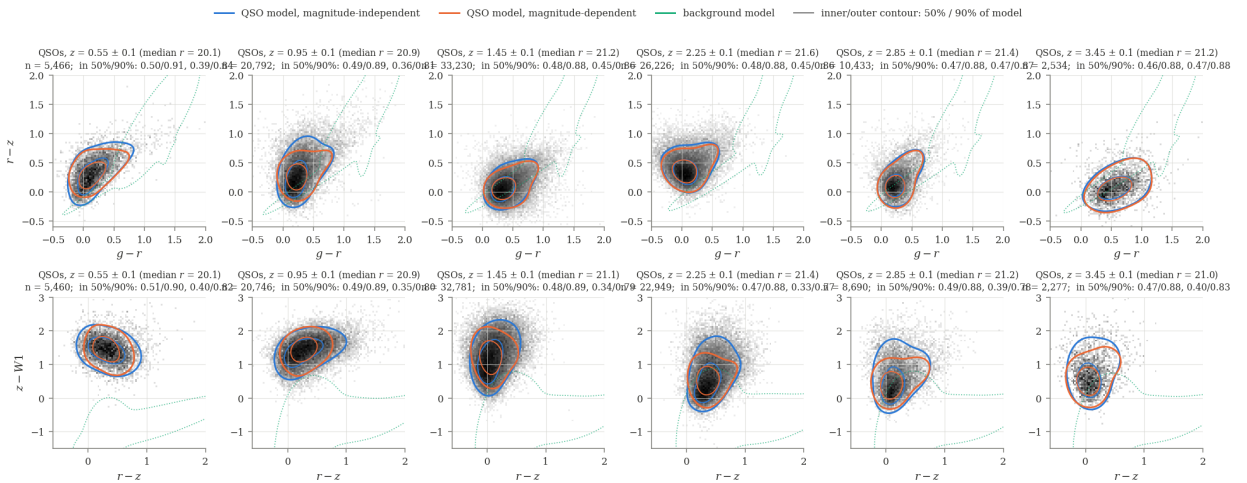


Figure 11: Quasar colour-colour planes in six redshift slices, each conditioned on the slice's median r , with the background model's 90% contour for reference.

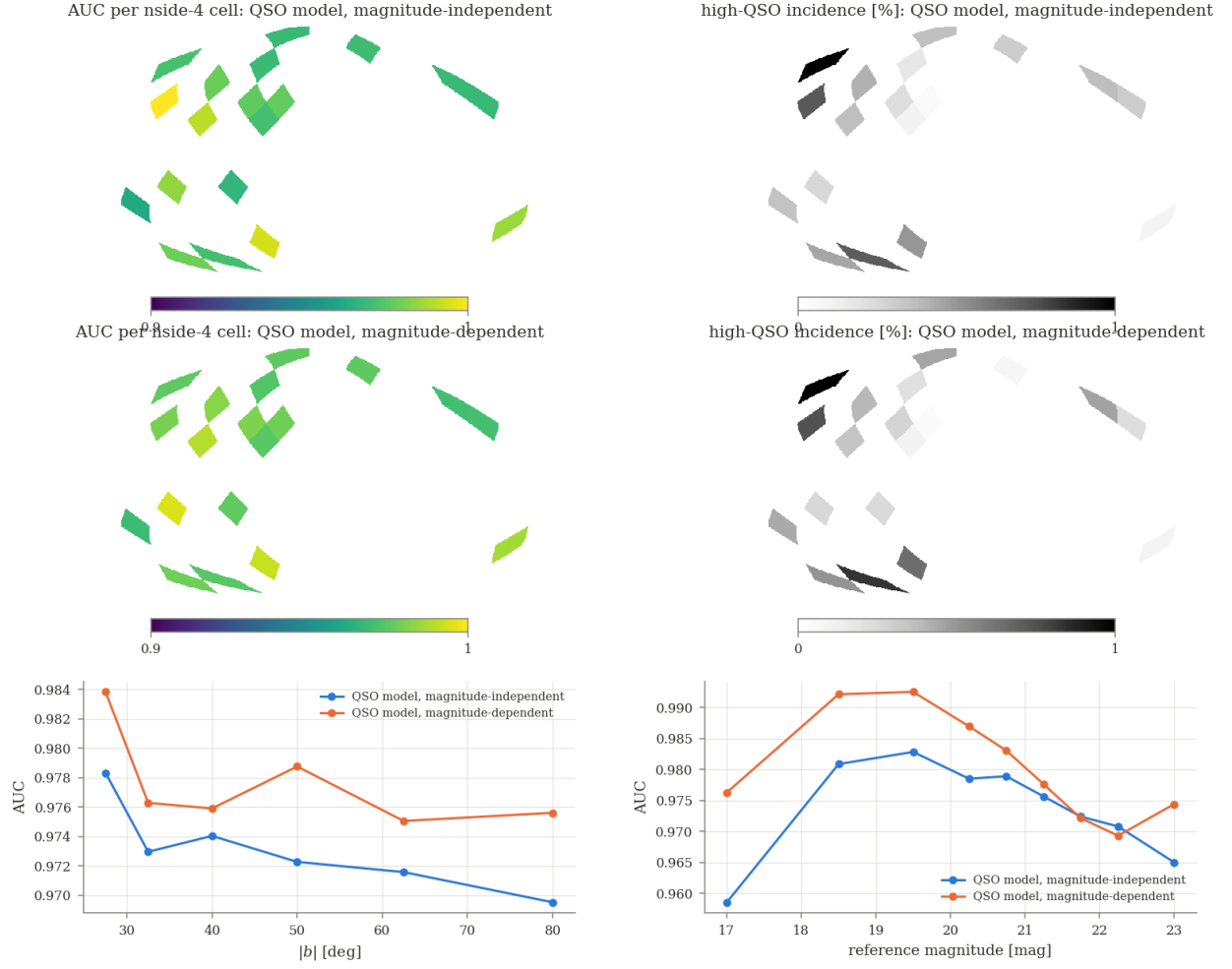


Figure 12: Performance against sky position and magnitude. Top: AUC and high-QSO incidence per HEALPix nside-4 cell (cells with at least 30 objects of each class). Bottom: AUC against $|b|$ and against reference-band magnitude.

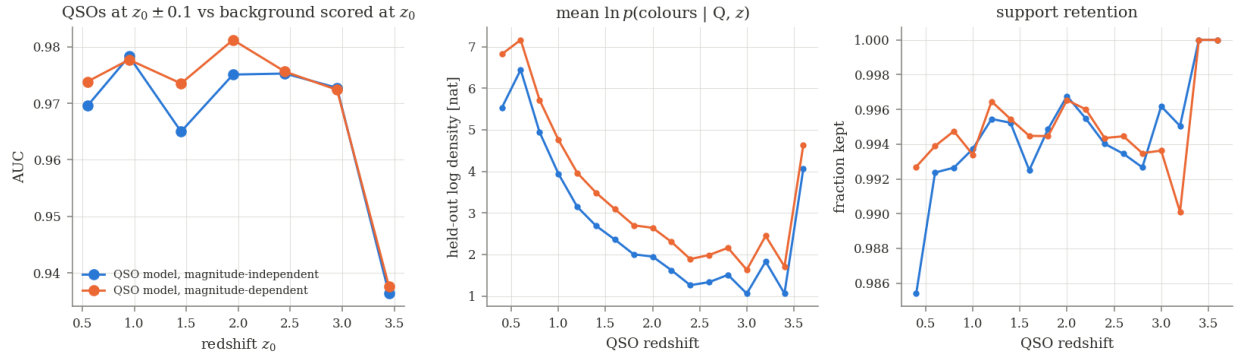


Figure 13: Performance against redshift. Left: AUC of quasars at $z_0 \pm 0.1$ against background objects scored at z_0 . Middle: mean held-out log density of quasars, in bins of their redshift. Right: support retention.

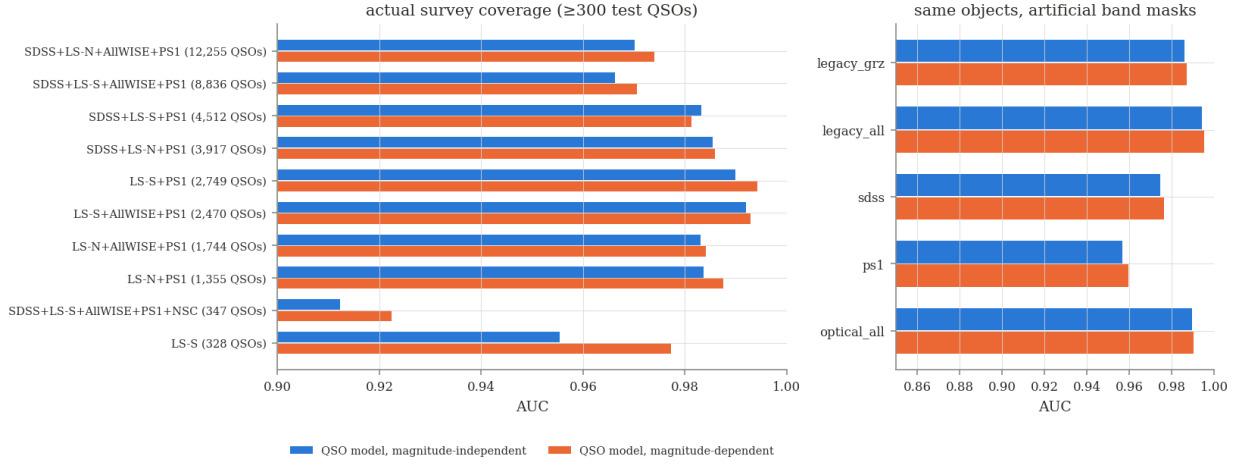


Figure 14: AUC by band coverage. Left: objects grouped by the surveys that observed them (groups with at least 300 test quasars). Right: the same objects scored with artificial band masks.

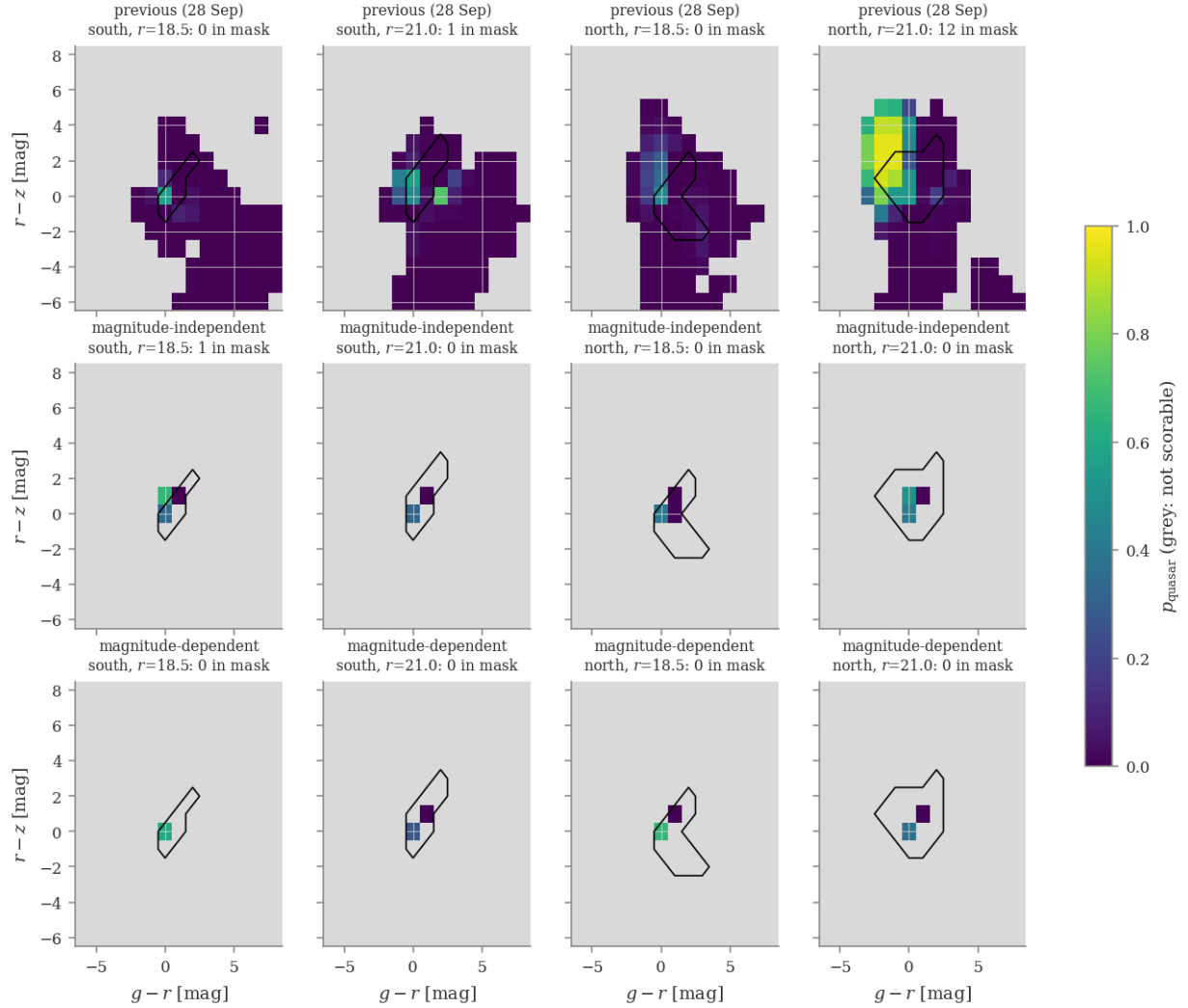


Figure 15: Total quasar probability on artificial Legacy colour grids, South and North, $r = 18.5$ and 21. Grey cells are not ranked. Black outline: the original low-density mask.

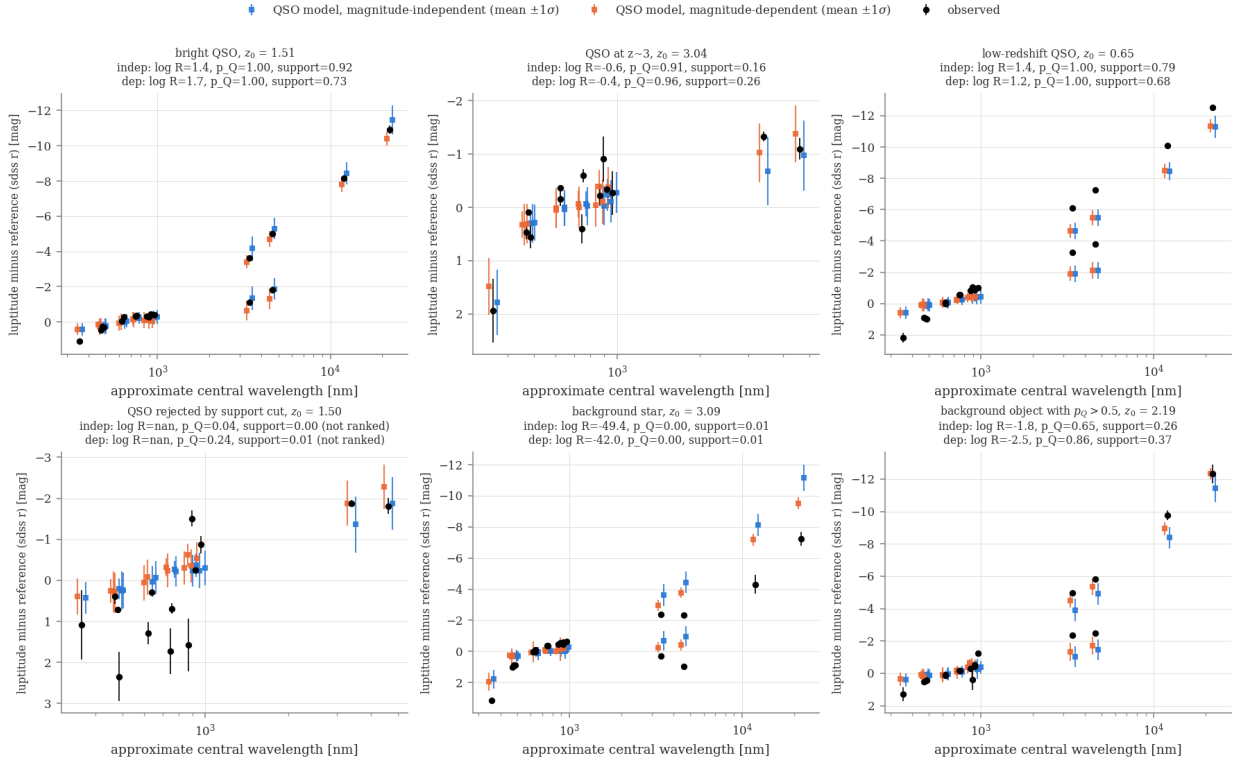


Figure 16: Worked examples: observed luptitude minus reference-band luptitude (black) and each model’s conditional prediction at the object’s redshift (mean $\pm 1\sigma$; blue magnitude-independent, or orange magnitude-dependent). Horizontal positions are approximate central wavelengths, for display only. Titles give $\log \mathcal{R}$, p_{quasar} and the support percentile from both models. Object identifiers are in `docs/method_unified/examples.json`.